



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to BRINDHAVAN FUEL AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

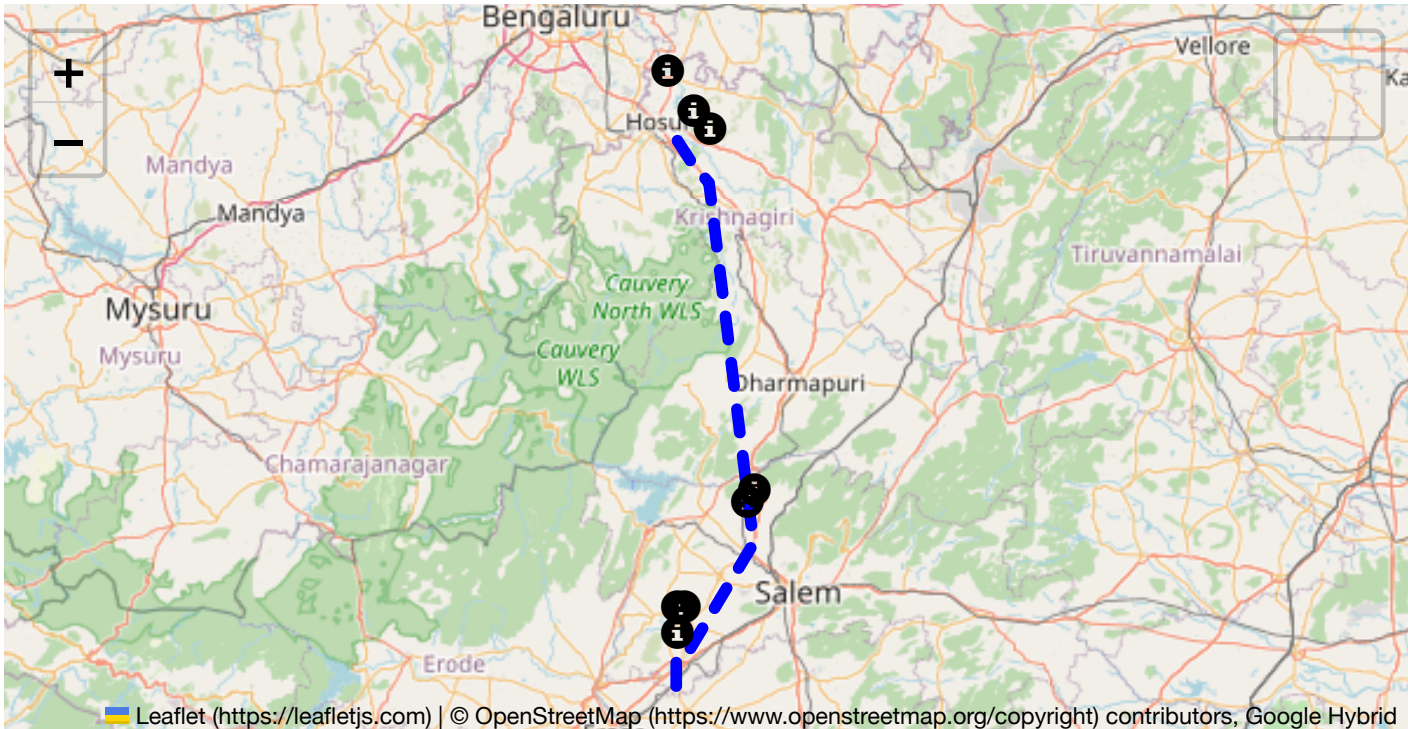
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 181.32 km
Estimated Duration: 3.8 hours
Adjusted Duration (Heavy Vehicle): 4.7 hours

Start: (11.4381, 77.8734)
End: (12.727299, 77.8512)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route begins in Sangagiri and proceeds to Hosur, covering key waypoints such as Devannagoundanur, Ayveli, Salem, Omalur, Kothur, and Thuppuganapalli. The majority of the route follows state highways with varying traffic conditions and road quality.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region typically experiences a tropical climate. During the summer months (March to June), high temperatures can affect vehicle performance, particularly tire pressures. Monsoon season (June to September) brings heavy rains that can cause flooding and landslides, particularly affecting areas around Salem with its hilly terrain. Fog can be prevalent in winters (November to February), reducing visibility.

3. Analysis of Traffic Patterns

- Sangagiri to Salem:** Moderate traffic, with peak hours in the morning (8 AM - 10 AM) and evening (5 PM - 8 PM).
- Salem City Area:** High congestion, particularly during weekdays and weekends in commercial zones.
- Salem to Omalur:** Busy during peak hours; potential delays can occur.
- Omalur to Hosur:** Variably busy with industrial traffic playing a vital role, especially near Hosur industrial areas.

4. Assessment of Road Quality and Infrastructure

- Highways:** Generally, in good condition but may have occasional potholes and uneven surfaces, particularly post-monsoon.

- **Rural Roads:** Some segments may lack proper signage and lighting, requiring extra caution at night.

5. Suggestions for Alternative Routes for Emergencies

- **If heavy congestion or accidents occur in Salem,** consider bypassing through peripheral roads like the Salem Bypass.
- **For weather-related delays,** SH 17 provides an alternative north-south route that could be used during emergencies.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials through Tamil Nadu requires permits and adherence to specific routes approved by regulatory authorities. Restricted timings often apply, usually limiting transit through urban areas during daytime hours due to safety and traffic concerns.

7. Overview of Historical Incidents

Historically, incidents in this region have often involved overturned trucks and spillage during monsoon due to slippery roads. Salem and Hosur have recorded accidents involving industrial chemicals due to poor road conditions and driver error.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas:** Consider the reserved forests near Hosur and Salem, where wildlife crossing can be a hazard.
- **Water Bodies:** Precaution is needed around rivers and lakes during monsoon to avoid waterlogging.

9. Analysis of Communication Coverage

The majority of the route has robust mobile network coverage, but rural stretches between Omalur and Kothur may experience weak signals, affecting communication reliability for several kilometers.

10. Estimated Emergency Response Times

- **Salem:** Being a large city, emergency services are prompt, typically 15-25 minutes.
- **Rural Areas:** Response times can extend to 45 minutes to an hour, depending on proximity to the nearest service center.

11. Overall Summary of Risk Assessment

This route poses several manageable risks to truck drivers, primarily related to fluctuating traffic conditions, weather impact, and the quality of rural roads. Monitoring weather updates, ensuring vehicle readiness for extreme temperatures, and remaining vigilant around congested areas can significantly decrease transportation hazards. Maintaining clear communication with relevant regulatory bodies regarding hazardous materials and staying informed about local traffic conditions are critical for a safe journey.

Careful preparation and real-time monitoring using apps can mitigate most challenges, ensuring the safety of drivers and goods along this essential route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	Medium	11.43995, 77.87431	30 KM/Hr
2	Turn	High	11.44107, 77.87528	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	Medium	11.46765, 77.87037	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	High	11.48277, 77.87250	15 KM/Hr
9	Turn	High	11.49288, 77.86815	15 KM/Hr
10	Blind Spot	Blind Spot	11.49009, 77.85114	10 KM/Hr
11	Blind Spot	Blind Spot	11.49392, 77.88536	10 KM/Hr
12	Turn	High	11.49435, 77.88499	15 KM/Hr
13	Turn	Medium	11.55536, 77.89740	30 KM/Hr
14	Turn	Medium	11.56559, 77.90634	30 KM/Hr
15	Turn	Medium	11.57613, 77.90844	30 KM/Hr
16	Turn	High	11.58281, 77.90796	15 KM/Hr
17	Turn	High	11.58320, 77.90646	15 KM/Hr
18	Turn	Medium	11.59093, 77.90595	30 KM/Hr
19	Turn	Medium	11.59261, 77.90697	30 KM/Hr
20	Turn	Medium	11.63768, 77.94809	30 KM/Hr
21	Turn	Medium	11.64121, 77.95138	30 KM/Hr
22	Turn	Medium	11.64323, 77.95184	30 KM/Hr
23	Turn	Medium	11.64587, 77.95108	30 KM/Hr
24	Turn	Medium	11.67896, 77.96882	30 KM/Hr
25	Turn	Medium	11.70049, 77.97964	30 KM/Hr
26	Turn	High	11.70179, 77.97900	15 KM/Hr
27	Turn	Medium	11.71394, 77.99572	30 KM/Hr
28	Turn	Medium	11.71739, 78.00032	30 KM/Hr
29	Turn	Medium	11.71984, 78.01216	30 KM/Hr
30	Turn	Medium	11.72770, 78.02297	30 KM/Hr
31	Turn	High	11.73786, 78.03505	15 KM/Hr
32	Turn	High	11.73654, 78.03889	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
33	Turn	High	11.74257, 78.04708	15 KM/Hr
34	Turn	Medium	11.86915, 78.08382	30 KM/Hr
35	Turn	Medium	11.91266, 78.05852	30 KM/Hr
36	Turn	Medium	11.93365, 78.06387	30 KM/Hr
37	Turn	Medium	11.93746, 78.05331	30 KM/Hr
38	Turn	Medium	11.95531, 78.04698	30 KM/Hr
39	Turn	High	11.95854, 78.04990	15 KM/Hr
40	Turn	High	11.96015, 78.04945	15 KM/Hr
41	Turn	Medium	11.96266, 78.05115	30 KM/Hr
42	Turn	High	11.96433, 78.05113	15 KM/Hr
43	Turn	High	11.96542, 78.04938	15 KM/Hr
44	Turn	Medium	12.12036, 78.13284	30 KM/Hr
45	Turn	Medium	12.28293, 78.06782	30 KM/Hr
46	Blind Spot	Blind Spot	12.51393, 78.03186	10 KM/Hr
47	Turn	Medium	12.51379, 78.02933	30 KM/Hr
48	Turn	Medium	12.51624, 78.02577	30 KM/Hr
49	Turn	Medium	12.52904, 78.02046	30 KM/Hr
50	Turn	Medium	12.67928, 77.86189	30 KM/Hr
51	Turn	Medium	12.68951, 77.85534	30 KM/Hr
52	Turn	Medium	12.69438, 77.85420	30 KM/Hr
53	Turn	Medium	12.71859, 77.83783	30 KM/Hr
54	Blind Spot	Blind Spot	12.71959, 77.83552	10 KM/Hr
55	Turn	Medium	12.72320, 77.83956	30 KM/Hr
56	Turn	Medium	12.72551, 77.84041	30 KM/Hr
57	Turn	Medium	12.72679, 77.84242	30 KM/Hr
58	Turn	High	12.72823, 77.84284	15 KM/Hr
59	Blind Spot	Blind Spot	12.73274, 77.83603	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
7	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
8	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
9	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
10	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
11	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
12	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
13	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
14	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
15	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
16	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
17	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
18	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
20	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
21	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
22	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
23	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
24	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
25	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
26	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
27	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
30	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
31	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
32	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium
36	pharmacy	H2 Pharmacy	12.7301802, 77.8401037	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
19	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
28	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
33	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium
34	school	Panchayat Union Primary School	12.6078768, 77.9381865	30 km/h	Medium
35	school	Panchayat Union Primary School	12.6835154, 77.8604798	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44107, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48277, 77.87250



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49288, 77.86815



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49009, 77.85114



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49392, 77.88536



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49435, 77.88499



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55536, 77.89740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58320, 77.90646



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59093, 77.90595



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63768, 77.94809



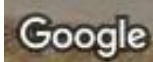
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64121, 77.95138



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64323, 77.95184

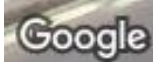


Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64587, 77.95108



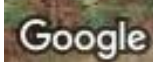
Risk Level: Medium

Coordinates: 11.67896, 77.96882



Risk Level: Medium

Coordinates: 11.70049, 77.97964



Risk Level: High

Coordinates: 11.70179, 77.97900



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71739, 78.00032



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71984, 78.01216



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72770, 78.02297



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73786, 78.03505



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73654, 78.03889



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.74257, 78.04708



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86915, 78.08382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93365, 78.06387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96266, 78.05115



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12036, 78.13284



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.28293, 78.06782



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.51393, 78.03186



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51379, 78.02933



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51624, 78.02577



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52904, 78.02046



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.67928, 77.86189



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.68951, 77.85534



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.69438, 77.85420



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.71859, 77.83783



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.71959, 77.83552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.72320, 77.83956



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.72551, 77.84041



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.72679, 77.84242



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.72823, 77.84284



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.73274, 77.83603

